

Recommendation & Feedback Letter —

PharmaNet Internship

 This document is part of the pharmanet-org/SRS documentation repository.

 Live deployments: [Admin Portal](#) · [Help Center](#) · [Mobile APK](#)

 Source code: github.com/pharmanet-org

1. Internship Experience

Team Introduction

We, the PharmaNet development team, completed our internship at **Alyah Software** building a comprehensive verified pharmacies marketplace platform from February through June 2026.

Team Members & Roles:

Name	Role	Primary Contributions
Belaynesh Fekadu	Software Tester, Documentation Maintainer	QA testing, user guide documentation (English + Amharic), content structuring, bug reporting
Bilal Worku (Bilal W. Suleyman)	Team Lead, Software Engineer	Architecture design, Flutter mobile app development, web portal development, CI/CD, project coordination
Yohannes Adane	Software Engineer, Supabase Database Administrator	Database schema design, RLS policies, API layer, real-time features, Chapa payment integration, Prisma ORM

Project Timeline & Key Milestones

Month	Milestone
February 2026	Project initialization — Flutter scaffold, Supabase project creation, initial auth setup
March 2026	Authentication system, product catalog with database-driven data, pharmacy registration flow
April 2026	Order management with real-time tracking, Chapa payment gateway, customer-pharmacist chat, English/Amharic localization
May 2026	Monetization features (pharmacy subscriptions, featured product boosting), separation of admin and seller web portals
June 2026	Help center documentation, technical docs, chatbot with local RAG, final testing, deployment, and demo delivery (admin portal live at web.pharmanet.bilsul.com , docs at pharmanet.bilsul.com/docs , mobile app v1.8.2 released on GitHub)

What We Built

Over 4 months, we delivered **6 production-ready projects** forming a complete platform:

1. **Flutter Mobile App** ([pharmanet/](#)) — 399 commits, v1.9.5, Android/iOS/Web/Desktop with 20+ features
2. **Admin Web Portal** ([pharmanet-admin/](#)) — 224 commits, full platform management dashboard
3. **Seller Web Portal** ([pharmanet-web/](#)) — 228 commits, pharmacy operations dashboard
4. **Help Center** ([pharmanet-guide-docs/](#)) — 55 commits, bilingual (English + Amharic), Mintlify
5. **Technical Documentation** ([docs/](#)) — 70+ pages, Zensical static site
6. **Support Chatbot** ([chatbot/](#)) — Local RAG with ChromaDB and ONNX embeddings

Technologies Mastered

- **Flutter/Dart** with Riverpod state management, Freezed data classes, clean architecture
- **React/Vite** with MUI 5, Redux Toolkit, Prisma ORM, and extensive Vitest test suites

- **Supabase** (PostgreSQL 15, Auth, Realtime WebSockets, Storage, Row-Level Security)
 - **Chapa** payment gateway integration (ETB, hosted checkout)
 - **Firebase Cloud Messaging** for push notifications with FCM token management
 - **Mintlify/Zensical** for documentation site generation
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2. Problems Identified & Fixed

Throughout development, we encountered and resolved numerous issues. Below is a categorized summary extracted from our commit history.

2.1 Mobile App (Flutter) — Bugs Fixed

UI & Rendering Issues

- **Sliver layout crash** (Apr 27): Resolved critical `SliverList` + `SliverGrid` layout crash on home page that caused white screening. Fixed by converting `GridView` to `SliverGrid` for web stability.
- **White screen on pending page** (Apr 28): Navigation bug causing blank screen on the pending orders page. Resolved navigation logic in order status routing.
- **Duplicate app bars** (Apr 3): Cleaned up duplicate `AppBar` instances in pharmacist detail pages and centered empty-state button.
- **Infinite rebuild loops** (Apr 27): Fixed `ChangeNotifier` infinite rebuild loops in pharmacist providers by adding proper state guards.
- **State updates during build frame** (Apr 27): Prevented `setState` during build phase in `ProfileNotifier` and `LocationNotifier`.

Firebase & Initialization

- **No Firebase App error on Web** (Apr 25): Web platform was not initializing Firebase correctly. Resolved by using `DefaultFirebaseOptions.currentPlatform` instead of hardcoded Android options.
- **Splash screen hang** (Apr 26): Made initialization resilient with error handling to prevent splash screen hanging during Firebase/Supabase init failures.
- **Multiple compilation errors** (Apr 25): Resolved 10+ compilation errors after Riverpod API changes and migrated legacy providers to modern `NotifierProvider` pattern.

Payment & Orders

- **Chapa page routing bugs** (May 20): Fixed incorrect route handling in Chapa payment callback and

success pages.

- **Order navigation + status display** (May 14): Fixed broken navigation from order list to detail page and incorrect status labels.
- **Cart state sync** (Apr 20): Refined cart state synchronization across the checkout flow to prevent stale data.

Chat System

- **Real-time chat UI** (Jun 4): Fixed chat UI to render incoming messages immediately via Supabase Realtime subscription instead of requiring manual refresh.
- **Chat room creation bug** (Apr 11): Product inquiry from an existing pharmacy chat was not creating a new room correctly. Fixed room creation logic.
- **New room navigation** (Apr 11): Navigation to newly created chat rooms was broken after room creation.

Authentication

- **Broken auth check for returning users** (Apr 3): Onboarding was re-showing on every launch because authentication state check was defaulting to false. Fixed the root auth provider logic.
- **Missing parenthesis syntax error** (Apr 25): Restored missing closing parenthesis in AuthPage social buttons that caused a syntax error.

API & Data

- **Missing `dart:typed_data` import** (Apr 27): Added missing import for web-compatible `Uint8List` handling in `PharmacyApi`.
- **Image parsing in `ProductModel`** (Apr 27): Implemented robust image URL parsing with fallback for images stored as strings vs arrays in Supabase.
- **Featured products JSON parsing** (Mar 31): Fixed `FeaturedProductProvider` to correctly parse JSON responses from Supabase.
- **Pharmacy seller profile JSON** (Mar 31): Fixed `PharmacySellerProfileProvider` to consume API responses with correct field mapping.

2.2 Web Portals (React) — Bugs Fixed

Supabase Integration

- **Ambiguous column references** (Apr 27): Resolved column reference `"id" is ambiguous` errors in queries joining `profiles` and `sellers` tables. Fixed by adding explicit table aliases throughout the service layer.
- **PostgreSQL timezone errors** (Apr 25): Standardized date handling with explicit `toUTCString`

conversion to prevent timezone mismatch between app server and database.

- **CMS banner field mapping** (Apr 25): Corrected field names between Supabase schema and UI form — status field was mapped incorrectly, causing all banners to show as "inactive".
- **Banner upload failure** (Apr 25): Uploads failing because storage bucket was created in a different region. Switched to verified bucket path with correct permissions.

Payments & Transactions

- **Chapa page routing** (May 20): Fixed routing in payment callback pages where incorrect URL parameters caused blank screens.
- **Transaction query aliases** (Apr 25): Updated all payment transaction queries with consistent table aliases to resolve ambiguous join errors in reports.
- **Analytics data binding** (Apr 25): `ProductAnalytics` page was binding chart data to wrong field names (used `units_sold` instead of `sold_count`).

UI & State

- **MUI Menu Fragment error** (Apr 25): Material-UI `MenuItem` must be a direct child of `Menu`. A `Fragment` wrapper was causing console errors and menu not opening. Restructured the header component.
- **Redux store missing exports** (Apr 2): After refactoring `userSlice.js`, several thunks and selectors were missing from exports, breaking user management pages. Restored all exports.

2.3 Patterns Identified

Issue Pattern	Occurrences	Resolution
Supabase ambiguous joins	5+ instances	Added explicit table aliases in all SELECT queries
Postgres timezone mismatches	3 instances	Standardized to UTC with <code>.toISOString()</code> conversion
Riverpod infinite rebuilds	4 instances	Added <code>keepAlive: true</code> and <code>shouldRebuild</code> guards
Firebase platform differences	2 instances	Used <code>DefaultFirebaseOptions.currentPlatform</code>
MUI component nesting	2 instances	Removed intermediate wrappers in menu/list structures

3. Comments on Our Mentor/Advisor — Ermias

Our advisor **Ermias** at Alyah Software was an exceptional mentor whose professionalism, patience, and dedication made a profound impact on our team and the project. From day one, he treated us not as interns but as colleagues, giving us the trust and responsibility to own significant parts of the platform while always being there to guide us when we needed help.

The way Ermias communicated with us — always respectful, always constructive, always pushing us to think bigger — is something we carry with us as a model for how a leader should work with a team. He celebrated our wins, helped us learn from our mistakes without judgment, and genuinely cared about our growth as professionals. We cannot thank him enough for the time, energy, and wisdom he invested in us.

SRS Review & Scope Correction

One of the most impactful moments was Ermias's thorough review of our initial Software Requirements Specification. He identified two critical scope issues that fundamentally reshaped the project:

1. **Web Dashboard Scope Correction:** We had initially listed the Web Dashboard as "Out of Scope," planning to focus only on the mobile app. Ermias pushed back: *"Remember that this project requires both a Mobile and a Web app. Right now, you have listed the Web Dashboard as 'Out of Scope.' We need the Web App for the Internal Admins to manage data and verify licenses. Please bring the Web App back into the scope!"* This was a turning point — we immediately brought the admin and seller web portals into scope, which became two of our most substantial deliverables (450+ combined commits).
2. **B2C (Citizen/Public User) Focus:** Ermias also redirected our focus from purely B2B to include the end customer: *"You've focused on the Business-to-Business (B2B) side, which is great, but let's add the Citizen/Public User. A regular person should be able to use the mobile app to search for a medicine and see which pharmacy has it in stock."* This feedback drove the entire customer-facing mobile app experience — product search, pharmacy directory, cart, orders, and chat.

Ongoing Guidance & Feedback Loop

Ermias maintained a consistent and professional feedback loop throughout the project. He was always reachable, always responsive, and always made time for us despite his own busy schedule.

- **Personal Connection:** Ermias never treated us as just workers. He checked in on us personally — after holidays he wrote: *"Hi Ermi. How was holidays? I hope you spent it well with family and*

belongings. I hope it was a blessed one!" He celebrated our progress: *"We made huge progress on our work in PharmaNet. We talked to a bunch of pharmacies. We got ideas from them and we are ready to show a demo soon hopefully."* His response: *"I really appreciate your team's commitment, especially in communicating with pharmacies; that's how real impact is created. I'm looking forward to seeing your continued progress, and I'm excited about the impact we can achieve together with PharmaNet."* This kind of encouragement meant the world to us.

- **Progress Check-Ins:** Regularly asked detailed questions about feature progress: *"How is your project going on? Is there any new feature you implement or existing feature you enhanced, like internal pharmacies management or ..."* These check-ins kept us accountable and focused on delivering tangible value.
- **QA & Bug Tracking Encouragement:** When we shared our Excel-based QA tracking sheet documenting payment integration and customer flow bugs, Ermias acknowledged the systematic approach and encouraged us to continue rigorous testing.
- **Live Demo Demands:** Ermias pushed us to deliver working software: *"As much as possible just build your mobile app and you can send it as APK and host the web also (provide the link). And I will see your demo live. Don't worry about payment integration or other extra features, just we need a demo."* This practical advice taught us to prioritize working demos over perfection.
- **Deployment & Delivery:** We delivered on his request within days — admin portal at `web.pharmanet.bilsul.com`, documentation at `pharmanet.bilsul.com/docs`, and mobile app release v1.8.2 on GitHub.

Marketing & Growth

Ermias also supported our professional development beyond coding. When we expressed interest in learning marketing for our planned campaign: *"We also want to learn how marketing works. We will learn from the team as well,"* he responded positively: *"You're welcome Bilal. Kindly we are eager to learn and grow together."* His encouragement to use tools like After Effects, Davinci Resolve, and Canva for our marketing campaign showed his investment in our holistic growth.

Technical Direction

Beyond product guidance, Ermias provided critical technical mentorship:

- **Architecture Reviews:** His database schema review in week 3 caught normalization issues that would have caused performance problems as the product catalog grew.

- **Agent-Driven Development:** We shared our challenges with AI agent development (free tier limitations, daily/monthly query caps). Ermias understood the constraints and helped us navigate them.
- **Practical Delivery:** His insistence on delivering APK files and web links rather than waiting for perfect implementations taught us agile delivery principles.

Ermias's blend of strategic product thinking, technical guidance, and genuine care for our professional development made him an exceptional advisor. His feedback directly shaped the project's scope, architecture, and our growth as engineers.

4. Comments on Alyah Software

Company Culture

Alyah Software provided a professional yet collaborative internship environment. The flat hierarchy meant we could directly communicate with senior developers and decision-makers, significantly accelerating our learning curve.

Work Environment

- Modern development tooling: VS Code, Git, GitHub Actions CI/CD, Supabase Studio
- Access to production-grade Firebase and Supabase projects with real data
- Regular stand-ups and sprint planning following agile methodology
- Focus on solving Ethiopian market problems with real-world healthcare impact

Sponsorship & Branding

PharmaNet is proudly sponsored by Alyah Software, Ethiopia. The app displays "Sponsored by Alyah Software, Ethiopia" on the splash screen (implemented in `lib/widgets/shared/splash_page.dart` line 49) and in localization files. The company's commitment to building local technology solutions for the Ethiopian healthcare sector was evident throughout the project.

Notable Practices

- Documentation treated as a first-class deliverable, not an afterthought — each feature required user-facing docs before merge
 - Bilingual support (English + Amharic) designed from day one, not retrofitted
 - Real device testing across Android and iOS devices, not just emulators
 - Security-first approach with database RLS policies reviewed before any production data access
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5. GitHub Repository Links

Project	Repository
Flutter Mobile App	https://github.com/pharmanet-org/pharmanet
Admin Web Portal	https://github.com/pharmanet-org/pharmanet-admin
Seller Web Portal	https://github.com/pharmanet-org/pharmanet-web
Help Center (Guide Docs)	https://github.com/pharmanet-org/pharmanet-guide-docs
Technical Documentation	https://github.com/pharmanet-org/technical
GitHub Organization	https://github.com/pharmanet-org

6. Full Name & Signatures

Name	Role	Signature
Belaynesh Fekadu	Software Tester, Documentation Maintainer	Belaynesh Fekadu
Bilal Worku (Bilal W. Suleyman)	Team Lead, Software Engineer	Bilal Worku
Yohannes Adane	Software Engineer, Supabase Database Administrator	Yohannes Adane

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